

Fundamental field	Symmetries	Decomposition into SO(3) irrep(s)	Source
$\mathcal{K}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}^{\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{2, 3}, GenSet[-(2,3)]]]	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi} - \frac{1}{2} \eta_{\alpha\chi} \mathcal{K}_{1^-}^{\#1}{}_{\beta} + \frac{1}{2} \eta_{\alpha\beta} \mathcal{K}_{1^-}^{\#1}{}_{\chi} + \frac{4}{3} \mathcal{K}_{2^-}^{\#1}{}_{\beta\chi\alpha} + \mathcal{K}_{1^+}^{\#2}{}_{\beta\chi} n_{\alpha} + \frac{1}{3} \eta_{\alpha\chi} \mathcal{K}_{0^+}^{\#1} n_{\beta} - \mathcal{K}_{1^+}^{\#1}{}_{\alpha\chi} n_{\beta} + \mathcal{K}_{2^+}^{\#1}{}_{\alpha\chi} n_{\beta} -$ $\frac{1}{2} \mathcal{K}_{1^-}^{\#1}{}_{\chi} n_{\alpha} n_{\beta} - \mathcal{K}_{1^-}^{\#2}{}_{\chi} n_{\alpha} n_{\beta} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{K}_{0^+}^{\#1} n_{\chi} + \mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} n_{\chi} - \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} n_{\chi} + \frac{1}{2} \mathcal{K}_{1^-}^{\#1}{}_{\beta} n_{\alpha} n_{\chi} + \mathcal{K}_{1^-}^{\#2}{}_{\beta} n_{\alpha} n_{\chi}$	$\mathcal{I}_{\alpha\beta\chi}$
SO(3) irrep	Symmetries	Expansion in terms of the fundamental field	Source SO(3) irrep
$\mathcal{K}_{0^+}^{\#1}$	Symmetry[0, $\mathcal{K}_{0^+}^{\#1}$, {}, StrongGenSet[{}, GenSet[]]]	$\mathcal{K}^{\beta}{}_{\alpha\beta} n^{\alpha}$	$\mathcal{I}_{0^+}^{\#1}$
$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}_0^{\#1\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{1, 2, 3}, GenSet[-(1,2), -(2,3)]]]	$\frac{1}{3} \mathcal{K}_{\alpha\beta\chi} - \frac{1}{3} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{3} \mathcal{K}_{\chi\alpha\beta} - \frac{1}{3} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{3} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} +$ $\frac{1}{3} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} + \frac{1}{3} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{1}{3} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} + \frac{1}{3} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{3} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta}$	$\mathcal{I}_{0^-}^{\#1}{}_{\alpha\beta\chi}$
$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{1^+}^{\#1\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{1^+}^{\#1}{}_{\alpha\beta}$
$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{1^+}^{\#2\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$\mathcal{K}_{\chi\alpha\beta} n^{\chi} + \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} - \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{1^+}^{\#2}{}_{\alpha\beta}$
$\mathcal{K}_{1^-}^{\#1}{}_{\alpha}$	Symmetry[1, $\mathcal{K}_{1^-}^{\#1\bullet 1}$, {●1→ -a}, StrongGenSet[{}, GenSet[]]]	$-\mathcal{K}^{\beta}{}_{\alpha\beta} + \mathcal{K}^{\chi}{}_{\beta\chi} n_{\alpha} n^{\beta} + \mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}_{1^-}^{\#1}{}_{\alpha}$
$\mathcal{K}_{1^-}^{\#2}{}_{\alpha}$	Symmetry[1, $\mathcal{K}_{1^-}^{\#2\bullet 1}$, {●1→ -a}, StrongGenSet[{}, GenSet[]]]	$\mathcal{K}_{\beta\alpha\chi} n^{\beta} n^{\chi}$	$\mathcal{I}_{1^-}^{\#2}{}_{\alpha}$
$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	Symmetry[2, $\mathcal{K}_{2^+}^{\#1\bullet 1\bullet 2}$, {●1→ -a, ●2→ -b}, StrongGenSet[{1, 2}, GenSet[(1,2)]]]	$-\frac{1}{2} \mathcal{K}_{\alpha\beta\chi} n^{\chi} - \frac{1}{2} \mathcal{K}_{\beta\alpha\chi} n^{\chi} - \frac{1}{3} \eta_{\alpha\beta} \mathcal{K}^{\delta}{}_{\chi\delta} n^{\chi} + \frac{1}{3} \mathcal{K}^{\delta}{}_{\chi\delta} n_{\alpha} n_{\beta} n^{\chi} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\chi} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\chi} n^{\delta}$	$\mathcal{I}_{2^+}^{\#1}{}_{\alpha\beta}$
$\mathcal{K}_{2^-}^{\#1}{}_{\alpha\beta\chi}$	Symmetry[3, $\mathcal{K}_2^{\#1\bullet 1\bullet 2\bullet 3}$, {●1→ -a, ●2→ -b, ●3→ -c}, StrongGenSet[{1, 2}, GenSet[-(1,2)]]]	$-\frac{1}{4} \mathcal{K}_{\alpha\beta\chi} + \frac{1}{4} \mathcal{K}_{\beta\alpha\chi} + \frac{1}{2} \mathcal{K}_{\chi\alpha\beta} - \frac{3}{8} \eta_{\beta\chi} \mathcal{K}^{\delta}{}_{\alpha\delta} + \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}^{\delta}{}_{\beta\delta} - \frac{3}{8} \mathcal{K}^{\delta}{}_{\beta\delta} n_{\alpha} n_{\chi} + \frac{3}{8} \mathcal{K}^{\delta}{}_{\alpha\delta} n_{\beta} n_{\chi} + \frac{1}{4} \mathcal{K}_{\beta\chi\delta} n_{\alpha} n^{\delta} + \frac{1}{2} \mathcal{K}_{\chi\beta\delta} n_{\alpha} n^{\delta} +$ $\frac{1}{4} \mathcal{K}_{\delta\beta\chi} n_{\alpha} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}^{\epsilon}{}_{\delta\epsilon} n_{\alpha} n^{\delta} - \frac{1}{4} \mathcal{K}_{\alpha\chi\delta} n_{\beta} n^{\delta} - \frac{1}{2} \mathcal{K}_{\chi\alpha\delta} n_{\beta} n^{\delta} - \frac{1}{4} \mathcal{K}_{\delta\alpha\chi} n_{\beta} n^{\delta} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}^{\epsilon}{}_{\delta\epsilon} n_{\beta} n^{\delta} + \frac{1}{4} \mathcal{K}_{\alpha\beta\delta} n_{\chi} n^{\delta} -$ $\frac{1}{4} \mathcal{K}_{\beta\alpha\delta} n_{\chi} n^{\delta} - \frac{1}{2} \mathcal{K}_{\delta\alpha\beta} n_{\chi} n^{\delta} + \frac{3}{8} \eta_{\beta\chi} \mathcal{K}_{\delta\alpha\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \eta_{\alpha\chi} \mathcal{K}_{\delta\beta\epsilon} n^{\delta} n^{\epsilon} - \frac{3}{8} \mathcal{K}_{\delta\beta\epsilon} n_{\alpha} n_{\chi} n^{\delta} n^{\epsilon} + \frac{3}{8} \mathcal{K}_{\delta\alpha\epsilon} n_{\beta} n_{\chi} n^{\delta} n^{\epsilon}$	$\mathcal{I}_{2^-}^{\#1}{}_{\alpha\beta\chi}$